Finding the Correct Visual Evidence Without Forgetting: Mitigating Hallucination in LVLMs via Inter-Layer Visual Attention Discrepancy

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Abstract

Large Vision-Language Models (LVLMs) have shown remarkable performance on a wide range of vision-language tasks. Despite this progress, they are still prone to hallucination, generating responses that are inconsistent with visual content. In this work, we find that LVLMs tend to hallucinate when they pay insufficient attention to the correct visual evidence and gradually forget it during the generation process. We empirically find that although LVLMs overall attend insufficiently to visual evidence, they exhibit sensitivity to the correct visual evidence in specific layers, with notable inter-layer discrepancy. Motivated by this observation, we propose a novel hallucination mitigation method that enhances visual evidence based on **I**nter-**L**ayer **V**isual **A**ttention **D**iscrepancy (**ILVAD**). Specifically, we obtain the attention weights from early generated tokens to visual tokens across layers and identify the tokens that are repeatedly activated as visual evidence, forming a saliency map. We then enhance attention to visual evidence during generation through the saliency map to reduce visual forgetting. In addition, we leverage the saliency map to obtain attention scores of generated text to visual evidence, in order to select and emphasize text tokens that are strongly grounded in visual evidence. Our method is training-free and plug-and-play. Multiple benchmark evaluations conducted on five recently released models show that our method can consistently mitigate hallucinations in different LVLMs over various architectures. Code is available at https://github.com/ytx-ML/ILVAD.

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1. Introduction

Large Vision-Language Models (LVLMs) (Liu et al., 2023; Zhu et al., 2024; Wang et al., 2024; Bai et al., 2025; Zhu et al., 2025) have achieved powerful cross-modal capabilities by aligning visual information with Large Language Models (LLMs), making significant progress in various vision-language tasks such as image captioning, visual question answering, grounded reasoning, and so on (Wang et al., 2023; Chen et al., 2025; Peng et al., 2024; Wang et al., 2026). However, LVLMs are still prone to hallucination, generating responses that are inconsistent with visual content. This issue seriously affects the reliability of LVLMs in high-risk areas such as medical diagnosis (Huang et al., 2025) and autonomous driving (Zhang et al., 2025a).

To mitigate hallucination, several studies introduce external knowledge bases for alignment training or fine-tuning (Zhao et al., 2023; Yu et al., 2024), but they incur significant computational and storage overhead. In comparison, recent studies have focused on train-free approaches during the inference stage. Among these, the decoding strategies (Leng et al., 2024; Kim et al., 2024; An et al., 2025; Wan et al., 2025; Huo et al., 2025) adjust the logits distribution during generation to reduce the influence of language priors. However, these methods operate primarily on the logits rather than the internal visual attention, thus failing to effectively promote LVLMs to focus on visual information. Some studies analyze hallucinations from the perspective of attention patterns. They observe that LVLMs tend to allocate excessive attention to visual tokens unrelated to the current query, e.g., background regions, referred to as visual attention sinks (Kang et al., 2025; Tang et al., 2025; Zhang et al., 2025b). Although reallocating attention can partially mitigate hallucinations, it remains limited when language priors remain dominant, as it does not genuinely increase the model's reliance on visual information.